COMPUTER COMMUNICATIONS & NETWORKS — QUIZ 1

Module: Lecture 2 Part 1 (Layered Architecture, TCP/IP, Addressing & Multiplexing)

Name: _____ Student ID: _____ Score: _____ / 24

1. Which of the following BEST describes the "divide-and-conquer" design philosophy in layered network architectures?

- ◯ a. Complex and unmanageable network communication tasks are partitioned into smaller, self-contained, and manageable tasks to reduce total design complexity.
- ◯ b. Each host device processes only its locally generated data without cooperating with intermediate routers.
- ◯ c. All physical and logical functions are merged into a single unified protocol layer to maximize processing speed.
- ◯ d. Communication lines transmit duplicate frames simultaneously across diverse paths to increase throughput.
- ◯ e. Devices eliminate protocol headers and rely solely on synchronized internal clocks.

2. What critical operational benefit does *Modularity* provide in a layered network architecture?

- ◯ a. It allows protocols or hardware at any individual layer to be modified, upgraded, or replaced independently without impacting the remaining layers.
- ◯ b. It guarantees that all network layers utilize an identical header format and addressing scheme.
- ◯ c. It completely eliminates the need for logical addressing at Layer 3 and Layer 4.
- ◯ d. It forces network administrators to update every layer of client devices at the same time.
- ◯ e. It reduces the total bandwidth consumed by headers to zero bits.

3. The TCP/IP model is frequently characterized as a "de facto standard." What does the term *de facto standard* denote?

- ◯ a. A standard formally codified and legally mandated by an international legislative treaty or governmental body.
- ◯ b. A standard that achieved universal acceptance through widespread commercial and practical adoption rather than government decree.
- ◯ c. A closed, proprietary specification requiring royalties and licensing fees prior to deployment.
- ◯ d. An experimental, short-term prototype protocol intended strictly for laboratory testing.
- ◯ e. A standard that has been superseded and deprecated by modern regulatory agencies.

4. During data encapsulation at a sending host, what does each layer add to the protocol data unit as it moves *downward* through the stack?

- ◯ a. A checksum trailer only, with no header.
- ◯ b. A full duplicate replica of the underlying application payload.
- ◯ c. Its own layer-specific header containing control information destined for its peer layer at the receiver.
- ◯ d. Nothing; the raw bitstream flows completely unchanged through all layers.
- ◯ e. A universal master IP header that overwrites all previous headers.

5. What happens to protocol headers during decapsulation at the receiving host?

- ◯ a. All headers are retained indefinitely and stored in the application's persistent database.
- ◯ b. Each layer inspects and removes the corresponding header added by its peer layer at the sender before passing payload upward.
- ◯ c. The physical layer strips all headers simultaneously before forwarding the pure payload to Layer 2.
- ◯ d. Headers are passed unchanged to the application layer, which parses all seven layers at once.
- ◯ e. Headers are swapped with local routing metrics and returned to the sender.

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6. When an IPv4 packet traverses multiple intermediate routers across the Internet from source host to destination host, which addresses mutate and which remain invariant?

- ◯ a. Source and destination MAC addresses change at each router hop, but original source and destination IP addresses remain constant throughout.
- ◯ b. Source and destination IP addresses change at each hop, while MAC addresses remain constant.
- ◯ c. Both MAC and IP addresses change at each intermediate router hop.
- ◯ d. Both MAC and IP addresses remain unchanged across the entire end-to-end path.
- ◯ e. Only the destination IP address changes at each hop; the source IP and MAC addresses stay identical.

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7. Which option correctly identifies the exact functional SCOPE of communication associated with OSI/TCP-IP Layers 2, 3, and 4?

- ◯ a. Layer 2: Process-to-process; Layer 3: Hop-to-hop; Layer 4: Host-to-host
- ◯ b. Layer 2: End-to-end; Layer 3: Host-to-host; Layer 4: Hop-to-hop
- ◯ c. Layer 2: Hop-to-hop (node-to-node); Layer 3: Host-to-host (source-to-destination); Layer 4: Process-to-process (end-to-end)
- ◯ d. Layer 2: Host-to-host; Layer 3: Process-to-process; Layer 4: Node-to-node
- ◯ e. Layer 2: Hop-to-hop; Layer 3: Process-to-process; Layer 4: Host-to-host

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8. What is the fundamental purpose of adding a header at an architectural layer?

- ◯ a. To compress the underlying application data so it occupies less bandwidth.
- ◯ b. To carry the control metadata, parameters, and addressing required for that layer to execute its functions.
- ◯ c. To physically boost the electrical voltage along the transmission medium.
- ◯ d. To eliminate the necessity of lower-layer protocols and hardware interfaces.
- ◯ e. To encrypt the entire transmission link with hardware-level security.

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9. Which statement correctly distinguishes Time Division Multiplexing (TDM) from Frequency Division Multiplexing (FDM)?

- ◯ a. TDM partitions transmission channel capacity into distinct frequency sub-bands, while FDM divides access by discrete time slots.
- ◯ b. TDM assigns different time slots to multiple channels over the full bandwidth, whereas FDM subdivides total channel bandwidth into separate frequency carriers.
- ◯ c. TDM applies exclusively to optical fiber media, whereas FDM applies exclusively to copper twisted-pair.
- ◯ d. TDM multiplexes packets based on IP addresses, while FDM multiplexes packets based on port numbers.
- ◯ e. TDM requires physical separate cables per user, whereas FDM requires only one cable.

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10. A standard IEEE 802 MAC address is 48 bits in length. How many hexadecimal digits are required to represent this address in human-readable format?

- ◯ a. 8 hexadecimal digits
- ◯ b. 12 hexadecimal digits
- ◯ c. 16 hexadecimal digits
- ◯ d. 24 hexadecimal digits
- ◯ e. 32 hexadecimal digits

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11. Which diagnostic command-line utility is used on Windows operating systems to inspect comprehensive TCP/IP network configuration details, including MAC address, IPv4 address, subnet mask, and default gateway?

- ◯ a.`ping -t`
- ◯ b.`tracert -d`
- ◯ c.`ipconfig /all`
- ◯ d.`netstat -rn`
- ◯ e.`nslookup -query`

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12. In the layered addressing hierarchy, which mapping correctly matches each address type to its precise standardized bit size?

- ◯ a. MAC address = 32 bits; IPv4 address = 48 bits; Port number = 16 bits
- ◯ b. MAC address = 48 bits; IPv4 address = 32 bits; Port number = 16 bits
- ◯ c. MAC address = 48 bits; IPv4 address = 64 bits; Port number = 32 bits
- ◯ d. MAC address = 64 bits; IPv4 address = 32 bits; Port number = 8 bits
- ◯ e. MAC address = 16 bits; IPv4 address = 32 bits; Port number = 48 bits

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13. An IPv6 address consists of 128 bits. When represented in standard colon-delimited hexadecimal notation, how many total hexadecimal digits comprise a full IPv6 address?

Show calculation: $\text{Hex digits} = \frac{\text{Total Bits}}{4}$

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14. Given that a Layer 4 Transport Layer port number is allocated exactly 16 bits, what is the maximum total number of distinct unique port numbers supported per transport protocol (e.g., TCP or UDP)?

Formula: $N = 2^k$ (where $k = 16$)

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15. In addition to a header, which OSI layer routinely appends a *trailer* (containing a Frame Check Sequence / CRC) to the encapsulated payload?

- ◯ a. Transport Layer (Layer 4)
- ◯ b. Network Layer (Layer 3)
- ◯ c. Data Link Layer (Layer 2)
- ◯ d. Session Layer (Layer 5)
- ◯ e. Application Layer (Layer 7)

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16. Match each Protocol Data Unit (PDU) designation to its associated architectural layer:

(1) Segment / Datagram → Layer name: _____

(2) Packet → Layer name: _____

(3) Frame → Layer name: _____

(4) Raw Bits → Layer name: _____

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17. Consider Host A sending a packet to Host B across Router R. Inside the Ethernet frame travelling over the link between Router R's outbound interface and Host B, which of the following is TRUE?

- ◯ a. The source MAC is Host A's MAC; destination MAC is Host B's MAC.
- ◯ b. The source MAC is Router R's outgoing interface MAC; destination MAC is Host B's MAC.
- ◯ c. The source IP is Router R's IP; destination IP is Host B's IP.
- ◯ d. Both source and destination MAC addresses are set to all F's (broadcast).
- ◯ e. IP addresses are completely replaced by MAC addresses in the packet header.

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18. Contrast "de facto" standards with "de jure" standards. Which of the following is the best representative example of a *de jure* standard?

- ◯ a. The QWERTY keyboard layout becoming dominant through market habit.
- ◯ b. An international communications standard officially approved and ratified by the ITU-T or ISO committees.
- ◯ c. A proprietary operating system protocol adopted because of overwhelming consumer software dominance.
- ◯ d. The TCP/IP suite emerging informally from ARPANET network deployments.
- ◯ e. A file formatting syntax popularized by a single independent software vendor.

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19. In Synchronous TDM, if a connected user has no active data to transmit during their allocated recurring time slot, what occurs?

- ◯ a. The time slot is immediately dynamically allocated to another active user.
- ◯ b. The time slot goes unused and travels empty, resulting in wasted transmission capacity.
- ◯ c. The entire multiplexer halts until the idle user resumes sending data.
- ◯ d. The multiplexer switches automatically into Frequency Division Multiplexing mode.
- ◯ e. The user is permanently disconnected from the frame schedule.

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20. What is a *Socket Address* in computer networking?

- ◯ a. The combination of a MAC address and an IP address.
- ◯ b. The combination of an IP address and a Port number (e.g.,`192.168.1.100:8080`).
- ◯ c. The physical RJ-45 wall outlet number in a wiring closet.
- ◯ d. The serial number burned into a network interface card (NIC).
- ◯ e. The URL string resolving a domain name into an IP address.

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21. Calculate the total transmission encapsulation overhead for an application message payload of 1,200 bytes that is wrapped with: a 20-byte TCP header, a 20-byte IPv4 header, and an Ethernet frame header (14 bytes) plus trailer (4 bytes). What is the total frame size transmitted across the physical medium?

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22. Which command-line utility tests reachability between two hosts by sending ICMP Echo Request messages and measuring Round-Trip Time (RTT)?

- ◯ a.`netstat`
- ◯ b.`ipconfig`
- ◯ c.`ping`
- ◯ d.`arp -a`
- ◯ e.`route print`

23. In the OSI 7-layer Reference Model, which layer is responsible for initiating, maintaining, synchronizing, and gracefully terminating communication sessions between applications?

- ◯ a. Transport Layer
- ◯ b. Session Layer
- ◯ c. Data Link Layer
- ◯ d. Presentation Layer
- ◯ e. Network Layer

24. Why are MAC addresses insufficient on their own for routing packets across the global Internet, thereby necessitating Layer 3 IP addresses?

- ◯ a. MAC addresses are flat and have no geographic or topological hierarchy, making global routing tables impossibly large.
- ◯ b. MAC addresses can only be processed by wireless routers and not copper or fiber connections.
- ◯ c. MAC addresses change every time an operating system reboots.
- ◯ d. MAC addresses cannot be expressed in binary or hexadecimal format.
- ◯ e. MAC addresses are only permitted on computers manufactured prior to 1995.

ANSWER KEY & DETAILED EXPLANATIONS

Lecture 2 Part 1 — Solutions

Question 1: a Divide and Conquer

Explanation: Layered architecture relies on the divide-and-conquer strategy: breaking down an overwhelmingly complex communication process into distinct, manageable functional layers with clearly defined boundaries[cite: 9].

Question 2: a Modularity

Explanation: Modularity allows developers and manufacturers to innovate or replace protocols within a given layer (e.g., swapping IPv4 with IPv6, or copper Ethernet with Wi-Fi) without needing to re-architect or rewrite applications above it[cite: 9].

Question 3: b Standards Terminology

Explanation: A *de facto* standard ("by fact") becomes an industry benchmark through broad adoption, widespread real-world deployment, and market consensus, rather than by being dictated by law or formal government mandates[cite: 10].

Question 4: c Data Encapsulation

Explanation: As data passes downward through the sending host's protocol stack, each layer encapsulates the data received from the layer above by prepending its own layer-specific header containing addressing and control data intended for its peer layer[cite: 10].

Question 5: b Decapsulation

Explanation: At the receiving end, the decapsulation process occurs in reverse: each successive layer parses and removes the corresponding header added by its peer layer at the sender before handing the remaining payload up to the next layer[cite: 10].

Question 6: a Hop vs End-to-End

Explanation: Data Link Layer MAC addresses operate hop-by-hop (node-to-node) and are rewritten across every router interface. Network Layer IP addresses represent end-to-end host addresses and remain unchanged from origin to final destination[cite: 11].

Question 7: c Layer Scopes

Explanation:

- **Layer 2 (Data Link):** Hop-to-hop (node-to-node between physically adjacent devices)[cite: 11].
- **Layer 3 (Network):** Host-to-host (source system to destination system across multiple intermediate hops)[cite: 11].
- **Layer 4 (Transport):** Process-to-process / End-to-end (delivering data to the specific executing software process/port)[cite: 11].

Question 8: b Header Purpose

Explanation: A protocol header carries control information (such as source/destination addresses, sequence numbers, checksums, and flags) required for that layer to coordinate with its remote peer layer and perform its defined functions[cite: 11].

Question 9: b Multiplexing Techniques

Explanation: Time Division Multiplexing (TDM) allocates the entire frequency band to different users during distinct, interleaved time slots. Frequency Division Multiplexing (FDM) divides the total available frequency bandwidth into distinct separate sub-channels for simultaneous transmission[cite: 12].

Question 10: b MAC Address Format
Explanation: Each hexadecimal character represents 4 bits ($2^4 = 16$). Therefore:

$$\text{Hex Digits} = \frac{48 \text{ bits}}{4 \text{ bits/digit}} = 12 \text{ hexadecimal digits}[\text{cite: } 12]$$

(Typically formatted as 6 pairs, e.g., `00:1A:2B:3C:4D:5E`).

Question 11: c Network Utilities
Explanation: On Microsoft Windows, `ipconfig /all` outputs detailed TCP/IP configuration parameters, including the physical (MAC) address, IP address, DHCP server, and DNS settings[cite: 12]. (Unix/Linux systems traditionally use `ifconfig -a` or `ip addr`).

Question 12: b Address Sizes
Explanation: Standard address bit lengths are:

$$\text{MAC Address (Layer 2)} = 48 \text{ bits } (6 \text{ bytes})[\text{cite: } 13]$$

$$\text{IPv4 Address (Layer 3)} = 32 \text{ bits } (4 \text{ bytes})[\text{cite: } 13]$$

$$\text{Port Number (Layer 4)} = 16 \text{ bits } (2 \text{ bytes})[\text{cite: } 13]$$

Question 13: 32 hexadecimal digits IPv6 Structure
Explanation: Since each hexadecimal digit encodes 4 bits:

$$\text{Hex Digits} = \frac{128 \text{ bits}}{4 \text{ bits/digit}} = 32 \text{ hexadecimal digits}$$

Organized into 8 groups of 4 hex digits separated by colons (e.g., `2001:0db8:85a3:0000:0000:8a2e:0370:7334`).

Question 14: 65,536 ports Port Range
Explanation: A 16-bit field allows:

$$N = 2^{16} = 65,536 \text{ unique port numbers}$$

Valid port numbers range from 0 to $65,535$ (Well-known: 0–1023, Registered: 1024–49151, Dynamic/Private: 49152–65535).

Question 15: c Trailers
Explanation: Layer 2 (Data Link Layer) adds both a header (source/destination MAC, type) and a **trailer** containing the Frame Check Sequence (FCS), typically computed using Cyclic Redundancy Check (CRC) to detect bit corruption during transmission over the physical link.

Question 16: PDU Names PDU Mapping
(1) Segment / Datagram: Transport Layer (Layer 4)
(2) Packet: Network Layer (Layer 3)
(3) Frame: Data Link Layer (Layer 2)
(4) Raw Bits: Physical Layer (Layer 1)

Question 17: b Router Framing
Explanation: When Router R forwards an IP packet onto the local segment connected to Host B, it strips the old Layer 2 frame and creates a brand-new Ethernet frame where the source MAC is Router R's egress interface and the destination MAC is Host B's physical interface[cite: 11]. The original IP source and destination remain untouched[cite: 11].

Question 18: b De Jure Standards
Explanation: A *de jure* standard ("by law" or formal regulation) is one that has been officially drafted, voted upon, and ratified by recognized standards bodies such as ISO, ITU, ANSI, or IEEE.

Question 19: b Synchronous TDM Waste

Explanation: In synchronous TDM, time slots are rigidly pre-assigned to channels regardless of whether they have data ready. If a host has nothing to send, its assigned time slot travels empty across the link, leading to inefficient bandwidth utilization. (Statistical TDM solves this by dynamically assigning slots on demand).

Question 20: b Socket Concept

Explanation: A network socket is the endpoint of a two-way communication link, uniquely identified by combining an IP address (identifying the specific host computer) and a Port number (identifying the specific software process on that host), written as `IP:Port`.

Question 21: Total size = 1,258 bytes (Overhead = 58 bytes) Encapsulation Overhead

Explanation:

$$\text{Overhead} = \text{TCP Header (20 B)} + \text{IP Header (20 B)} + \text{Ethernet Header (14 B)} + \text{Trailer (4 B)} = 58 \text{ bytes}$$

$$\text{Total Frame Size} = 1,200 \text{ B (Payload)} + 58 \text{ B (Overhead)} = 1,258 \text{ bytes}$$

Question 22: c Ping Utility

Explanation: `ping` uses ICMP (Internet Control Message Protocol) Echo Request and Echo Reply messages to verify IP-level connectivity and calculate round-trip packet latency between hosts.

Question 23: b OSI Session Layer

Explanation: The Session Layer (Layer 5) establishes, manages, synchronizes (checkpoints), and terminates conversations, dialogs, and sessions between remote software applications.

Question 24: a Flat vs Hierarchical

Explanation: MAC addresses are flat 48-bit hardware identifiers assigned by the vendor with no geographic or network topological hierarchy[cite: 13]. If the Internet routed on MAC addresses, every router in the world would have to store billions of individual MAC routes. IP addresses are hierarchically structured (network prefix + host ID), allowing massive route aggregation.